RRL - CHAPTER 2 (AI-Based Diet and Nutritional Planner)

Local Literature

1. Smith et al. (2024) conducted a scoping review of diets, fruit and vegetable intake, and nutritional status among children, adolescents, and adults in the Philippines. The review found that Filipino diets generally lacked diversity and were characterized by high consumption of energy-dense, nutrient-poor foods and low intake of fruits and vegetables. The authors emphasized the need for interventions that promote dietary diversity and healthier food choices. This literature supports the importance of improving eating practices and increasing access to nutrient-rich foods in the Philippine population.

Link:
https://scholar.google.com/scholar?as_ylo=2025&q=Nutrition+and+dietary+practices+in+the+Philippines&hl=en&as_sdt=0,5&scioq=artificial+intelligence+in+nutrition#d=gs_qabs&t=1791095178481&u=%23p%3DkT24gwvYL9EJ

2. Duay et al. (2026) discussed the applications of artificial intelligence (AI) in food science and nutrition, particularly its role in nutritional modeling, dietary assessment, and personalized nutrition planning. The authors explained that machine learning and deep learning techniques can improve the accuracy of nutritional profiling and assist in processing complex nutritional data. AI-based tools can also integrate different types of information, including dietary habits and other health-related data, to support personalized nutrition plans and predict individual health outcomes. The literature further highlighted the potential of AI to address nutrition-related concerns, improve dietary practices, and contribute to public health and food security. However, the authors also identified challenges such as the need for high-quality data and concerns regarding data privacy and ethics. This literature is relevant to the present research because it highlights the potential of AI to analyze nutritional information and provide personalized dietary recommendations, supporting the development of technology-assisted nutrition planning and healthier dietary practices.

Link:
https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&as_ylo=2025&scioq=artificial+intelligence+in+nutrition&q=ai+nutrition+in+philippines+literature&oq=ai+nutrition+in+philippines+litera#d=gs_qabs&t=1791097363914&u=%23p%3DgE0gv23BkfIJ

3. Tampol (2026) discussed the relationship between obesity, dietary patterns, micronutrient deficiencies, and inflammation among Filipino schoolchildren. The opinion paper reviewed findings from recent studies involving Filipino children in Metro Manila, which reported associations between overweight or obesity, micronutrient status, consumption of ultra-processed foods, and inflammatory biomarkers. The paper highlighted that Filipino children may experience the double burden of malnutrition, in which obesity and micronutrient deficiencies occur simultaneously. It also emphasized the possible contribution of frequently consumed ultra-processed foods, such as commercial baked goods, instant noodles, packaged cereals, and processed rice snacks, to inflammation risk. Furthermore, the author discussed the limitations of existing cross-sectional studies and proposed the gut microbiome-immunity relationship as a framework for future research on childhood

metabolic inflammation. This literature is relevant to the present research because it emphasizes the importance of considering both nutritional quality and dietary patterns when promoting healthy eating and addressing obesity and micronutrient deficiencies among Filipino children.

Link:
https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&as_ylo=2025&scioq=artificial+intelligence+in+nutrition&q=Philippine+nutrition+literature&btnG=#d=gs_qabs&t=1791097466362&u=%23p%3DhC3jRA1P0_oJ

4. Polii et al. (2025) reviewed the management of the Philippines' School-Based Feeding Program (SBFP) and examined lessons that could be applied to Indonesia's Makan Bergizi Gratis program. The authors analyzed the implementation and outcomes of the Philippine program, which was reported to have rehabilitated 62% of severely wasted children and improved school attendance. However, the review also identified challenges in monitoring and evaluation, with only 43% of beneficiaries being correctly verified, as well as concerns regarding the sustainability of nutritional improvements, since only 48% of children maintained normal nutritional status 12 months after the program. Based on these findings, the authors recommended stronger legal frameworks, improved logistics, comprehensive nutrition education, and digital monitoring and evaluation systems. This literature is relevant to the present research because it emphasizes the importance of nutrition programs, continuous monitoring, nutrition education, and sustainable approaches in improving the nutritional status of students.

Link: (SAME LINK 5)
https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&as_ylo=2025&scioq=artificial+intelligence+in+nutrition&q=nutritional+planner+in+Philippines+literature&btnG=#d=gs_qabs&t=1791097549453&u=%23p%3DQ-lPsWB_VDwJ

5. Dela Luna et al. (2025) examined the effect of capacity-building activities on the knowledge, attitudes, and self-efficacy of local agriculture planners regarding Nutrition-Sensitive Agriculture (NSA) in CALABARZON. The study used a non-randomized controlled trial design involving 57 municipal agriculturists and assessed changes in their knowledge, attitude, and self-efficacy after the capacity-building intervention. The results showed significant improvements in the participants' knowledge, attitudes, and self-efficacy regarding nutrition-sensitive agriculture. Membership in a local nutrition committee was also associated with greater improvements in knowledge and self-efficacy, while employment tenure was significantly related to changes in attitude. The researchers concluded that capacity-building activities can improve the ability of local agriculture planners to integrate nutrition-sensitive approaches into municipal agricultural development plans. This study is relevant to the present research because it demonstrates the importance of nutrition education and capacity-building in improving knowledge and practices related to nutrition and supports the development of strategies that promote healthier and more nutrition-sensitive communities.

Link:
https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&as_ylo=2025&scioq=artificial+intelligence+in+nutrition&q=nutritional+planner+in+Philippines+literature&btnG=#d=gs_qabs&t=1791097549453&u=%23p%3DQ-lPsWB_VDwJ